

URBAN RIG Demo Test Operation Test Report											One World Techno Co., Ltd.				
Experiment date and time	December 12, 2020		Weather	Sunny		Temperature	12°C		Start time	9:44		End time	5:52 PM		
Request destination	One World Co., Ltd.														
Experiment Manager	Sakano							Driver			Itakura and Ishikawa				
Equipment used	4.8 ton machine (No. 2) URB-50-2			Fuel used		Heavy oil A: approx. 280			Power consumption		Water usage				
Object	450kg of waste tires														
Driving location	One World Techno Okayama Head Office Factory														
Pyrolysis Furnace Operation Setting Details															
PR_0	Superheated steam		Pyrolysis Furnace			Boiler	Vapor SV5	Steam Burner B1	Heat Furnace Burner B2	Cooling Fan F1/F2	Suction Fan EP	Cooling Tower	MV1	MV2	PTL
STEP	Time	Temperature	Time	Temperature											
4	120	500	120	350	Driving	Open	Driving	Driving	Driving	Driving	Driving	Overheat	B	Green	
6	150	500	150	450	Driving	Open	Driving	Driving	Driving	Driving	Driving	Overheat	B	Green	
9	180	530	180	530	Driving	Open	Driving	Driving	Driving	Driving	Driving	Overheat	B	Green	
11	0	530	0	530	Driving	Open	Driving	Driving	Driving	Driving	Driving	Overheat	B	Green	
13	0	530	0	530	Driving	Open	Driving	Driving	Driving	Driving	Driving	Overheat	B	Green	
15	60	150	60	150	Stop	Close	Stop	Stop	Driving	Driving	Driving	Overheat	B	Yellow	
16	1	0	1	0	Stop	Close	Stop	Stop	Driving	Driving	Driving	Overheat	B	Yellow	

About processed materials

1. Photo before processing. Weight measurement



2. Photograph of the product placed inside the furnace before processing



3. Photograph of the product placed inside the furnace after processing



4. Photograph of the product placed inside the furnace after processing



Results and Discussion

450 kg of waste tires were pyrolyzed in stages. The waste tires were broken down into wire, charcoal, and oil.

The weight of each component after pyrolysis was 111 kg of wire, 133 kg of charcoal, and approximately 180 L of oil.

The next challenge is to shorten the processing time.

About the temperature graph



	T E 1	Steam temperature
	T E 2	Superheated steam outlet temperature
	T E 3	Superheated steam furnace inlet temperature
	T E 4	Combustion chamber temperature
	T E 5	Furnace lower temperature
	T E 6	Medium temperature inside the furnace
	T E 7	Furnace top temperature
	T E 9	Off-gas catalyst outlet temperature

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15	60	150	60	150	Stop	Close	Stop	Stop	Driving	Driving	Driving	Overheat	B	Yellow
16	1	0	1	0	Stop	Close	Stop	Stop	Driving	Driving	Driving	Overheat	B	Yellow

Regarding the photo after processing Weight after processing etc. Charcoal: 133 kg Metal (wire): 111 kg Oil: approx. 180 L

1. Condition inside the furnace after processing

2. Condition of oil discharged into the oil-water separator

3. After removal, during measurementWire1

4. After taking out, during measurementWire2

5. After removal, during measurementWire3

6. After taking out, during measurementWire4

7. After removal, during measurementWire5

8. After removal, during measurementWire6

9. After removal, during measurementWire7

10. After removal, during measurementWire8

11. Wire Measuring all at once

12. Measuring carbides together